



Midterm – MATH 346/377 – Winter 2025

DATE: Wednesday, February 26 TIME: 8:35 – 9:55

EXAMINER: Romain Branchereau TOTAL: 40 points

First name (please write as legibly as possible within the boxes)

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Last name

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Student ID number

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- There are four problems, each worth 10 points, for a total of 40 points. There is a fifth bonus problem, also worth 10 points.
- Notes, calculators and other electronic devices are not permitted.
- Write your solutions directly on the booklet, under the question (you can use the back of the page).
- The last two pages can be used as scrap paper and will not be graded. Please detach them before submitting.



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1. (10 points) True or false? Justify your answer.
 - (a) Let a, b, c be nonzero integers. Then $\gcd(a, b) + c = \gcd(a + c, b + c)$.
 - (b) The equation $319x + 30y = n$ has a solution $x, y \in \mathbb{Z}$ for any n .
 - (c) There are four elements of order 5 modulo 13.
 - (d) If $\gcd(n, m) = 1$, then $\gcd(\phi(n), \phi(m)) = 1$.



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2. Let $a \geq 1$ and $n \geq 2$ be two integers.

(a) (5 points) If $a^n - 1$ is prime, show that $a = 2$.

(b) (5 points) If $2^n - 1$ is prime, show that n is prime.

Primes of the form $2^p - 1$ are called Mersenne primes, and are not related to Fermat primes $2^{2^n} + 1$.



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3. (10 points) Find all integers $0 \leq x \leq 54$ that solve the equation

$$f(x) = 3x^{22} + 11x^5 + 8x^2 + 2x - 5 \equiv 0 \pmod{55}.$$



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4. (10 points) (a) (3 points) How many distinct primitive roots are there modulo 19?
- (b) (7 points) Let g be a primitive root modulo 19. Show that $g^6 \equiv 7 \pmod{19}$ or $g^6 \equiv 11 \pmod{19}$.



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5. (BONUS 10 points) Is the number

$$\alpha := \frac{1}{\sqrt{\sqrt{2} + \sqrt{3}}}$$

an algebraic *integer*? Is it a rational number?



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